

**From Person-Specific Dynamics to Population Inference: A Computationally Efficient
Meta-Analytic Structural Equation Modeling Framework for Integrating Complex Multilevel
Dynamic Models**

Links

Research Compendium

The data and materials for this study are available on OSF (<https://osf.io/uenr7>) and GitHub (<https://github.com/jeksterslab/manMetaVAR>, <https://jeksterslab.github.io/manMetaVAR/index.html>).

fitVARMxID R Package

Source code and documentation for the `fitVARMxID` R package are available on GitHub (<https://github.com/jeksterslab/fitVARMxID>, <https://jeksterslab.github.io/fitVARMxID/index.html>).

metaDyn R Package

Source code and documentation for the `metaDyn` R package are available on GitHub (<https://github.com/jeksterslab/metaDyn>, <https://jeksterslab.github.io/metaDyn/index.html>).

Single Replication from the Simulation Study

This vignette presents a complete walk-through of one representative Monte Carlo replication, including data generation, estimation for each competing method, and selected output and diagnostics. Convergence diagnostics for a representative BMLVAR (Bayesian multilevel vector autoregressive model) replication, including PSR, ESS, and trace plots, are provided via the `Mplus` output section of the simulation-replication vignette.

<https://jeksterslab.github.io/manMetaVAR/articles/sim-rep.html>

Empirical Illustration: Meta-Analysis of Affect Dynamics

This vignette reproduces the empirical example reported in the manuscript, including data preprocessing, person-specific VAR estimation, second-stage meta-analytic synthesis, and the resulting population-level affect-dynamics summaries.

<https://jeksterslab.github.io/manMetaVAR/articles/empirical.html>

Benchmarking MetaVAR Against an Uncertainty-Uncorrected Two-Stage Approach and Bayesian Multilevel VAR

This vignette reports the benchmarking analysis comparing the computational demands of MetaVAR, the uncertainty-uncorrected two-stage benchmark, and BMLVAR under a common simulation setting.

<https://jeksterslab.github.io/manMetaVAR/articles/benchmark.html>

Containers for Reproducibility

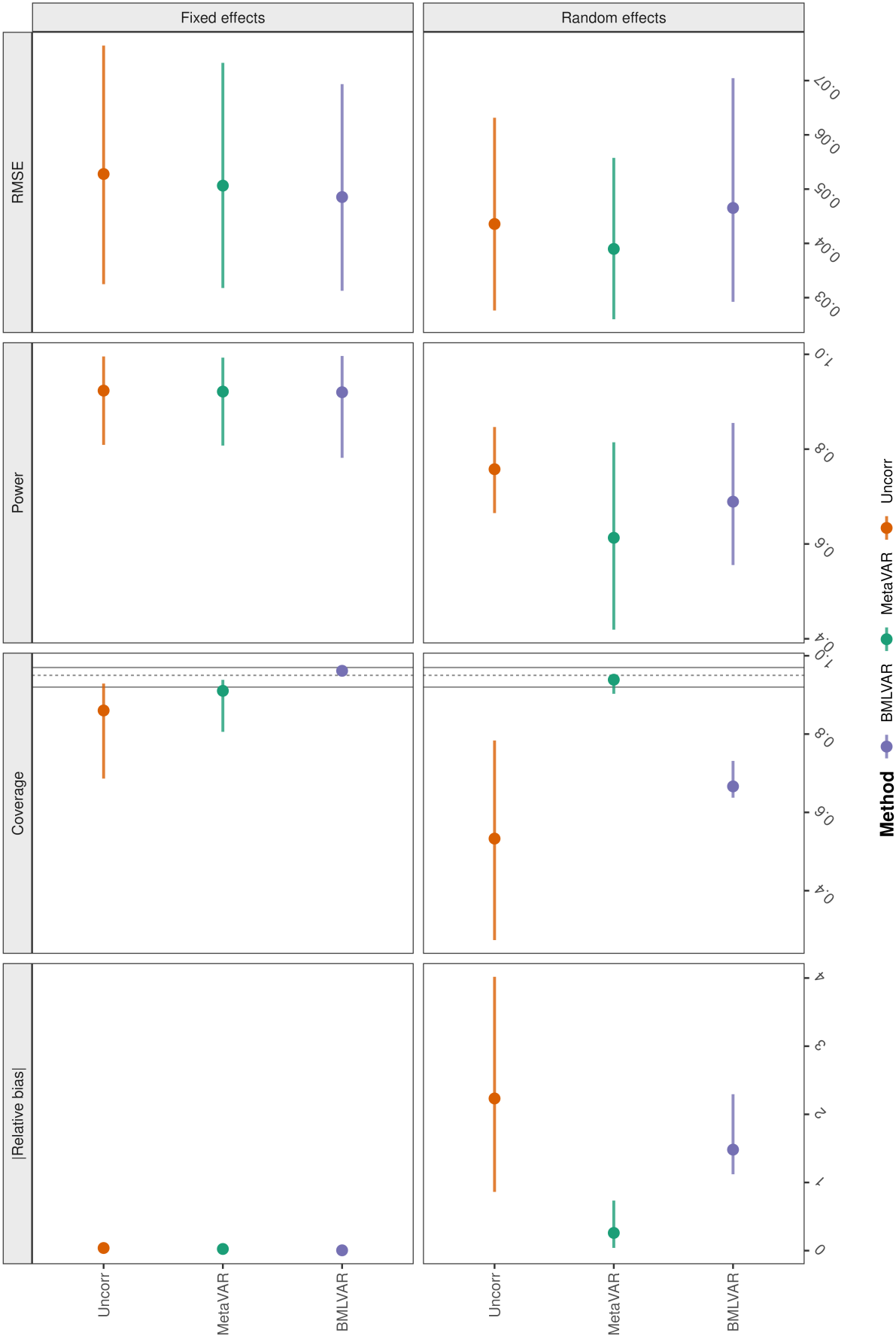
This vignette provides the containerized computing environments and step-by-step instructions needed to reproduce the analyses with the required software stack and dependencies.

<https://jeksterslab.github.io/manMetaVAR/articles/containers.html>

Figure 1
Overall Performance

Overall method performance across simulation design cells

Points are means across design cells; horizontal lines show the min-max range. Coverage panels include nominal .95 and Bradley's liberal .92-.97 band.



Coverage heatmap by parameter, design cell, and method

Negative values indicate undercoverage; zero indicates nominal coverage.

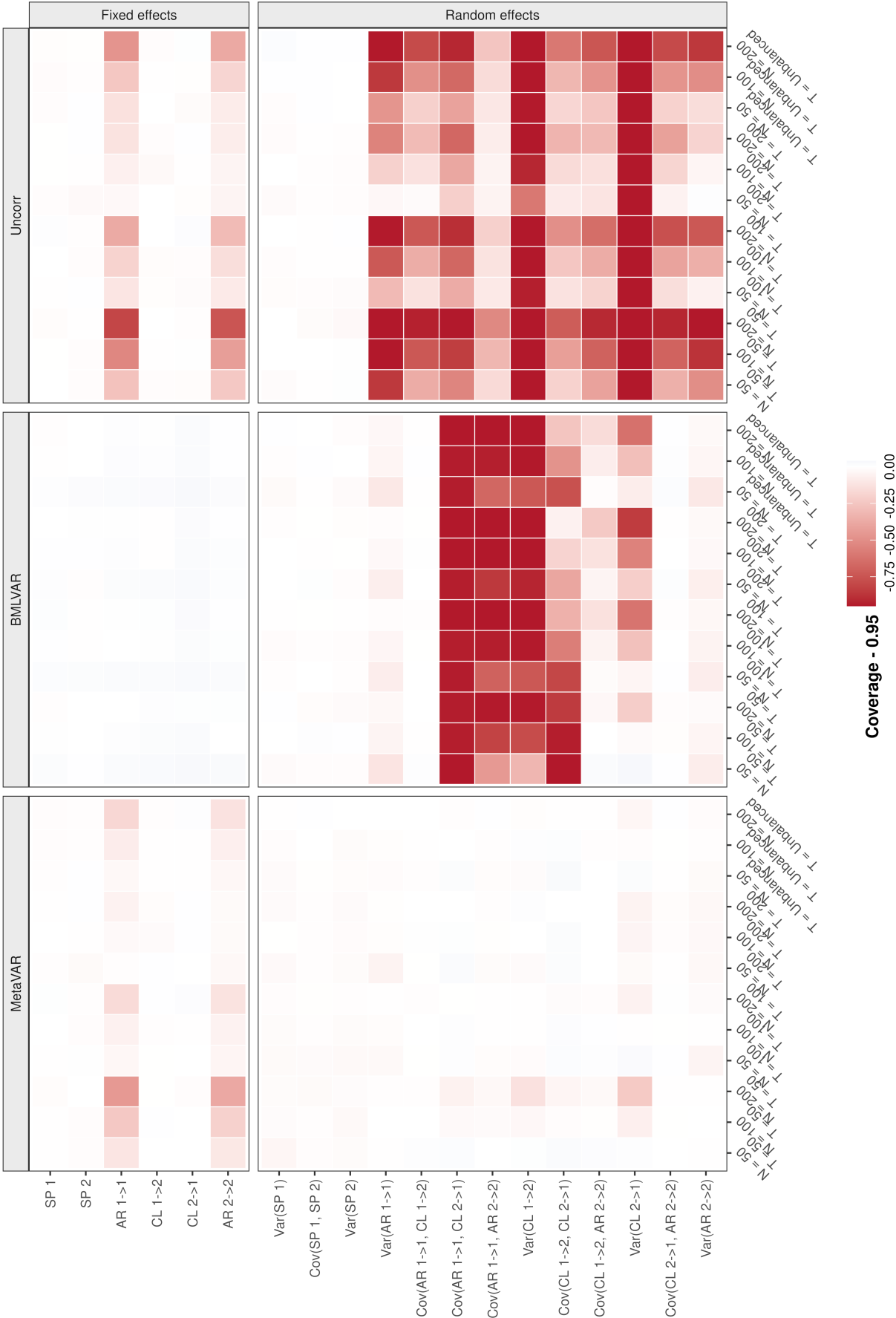


Figure 2
Coverage

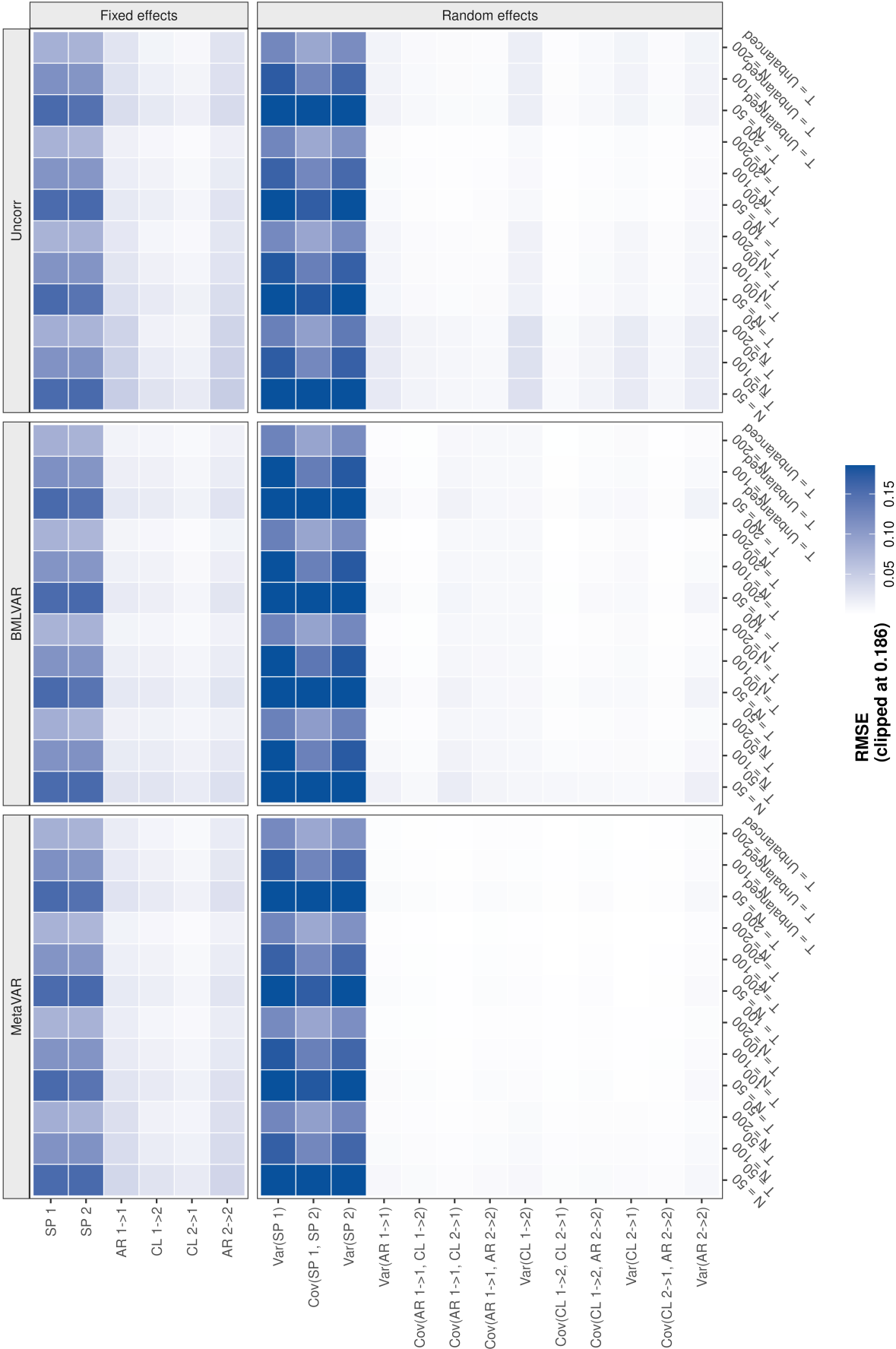
Absolute value of the relative bias heatmap by parameter, design cell, and method

Smaller values are better. The heatmap shows magnitude only, not direction.

Figure 4
RMSE

RMSE heatmap by parameter, design cell, and method

To keep the heatmap readable, RMSE is clipped at the 95th percentile.



Power heatmap by parameter, design cell, and method

